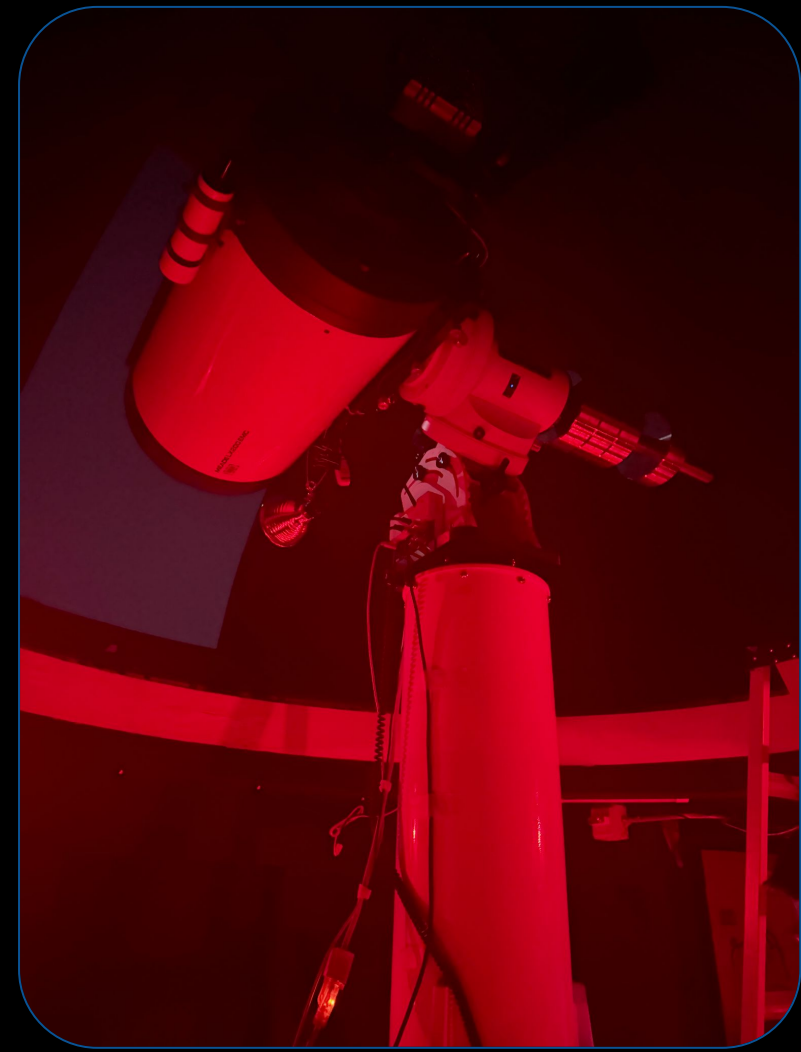


Old But Gold: Measuring the Age and Metallicity of M2 & M15 Star Clusters

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Abstract

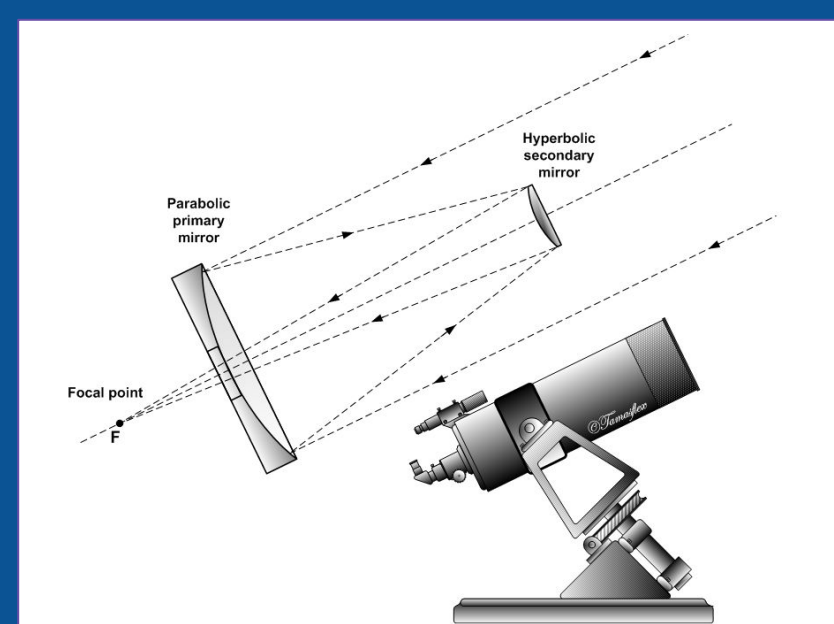
Star clusters are great laboratories for studying stars and their properties because all of the stars were born as a single population, thus having the same age, composition, and distance. Obtaining precise age measurements of star clusters is important to further constrain and test models of stellar and galactic evolution including, stellar dynamics, galaxy formation, and chemical evolution. Using UW-Madison's 16-inch Linda Sue optical telescope in the Livingston-Stark Observatory, we collected photometric data on the globular clusters Messier 2 (M2) and Messier 15 (M15) in multiple filters and measured each star's brightness. After calibration, we estimated the age of M2 and M15 to be 12.589 Gyr and 12.5 Gyr respectively, by fitting a stellar evolution model to our observed color-magnitude diagram.



Observations

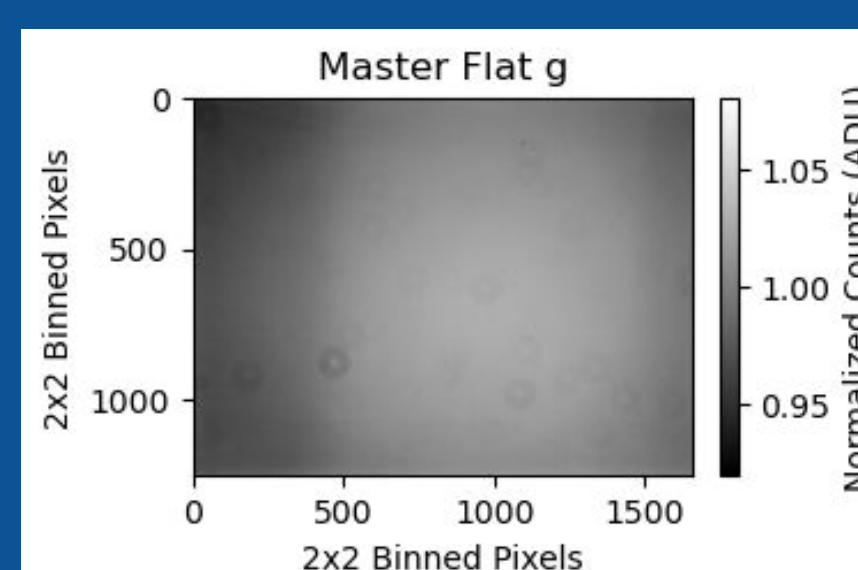
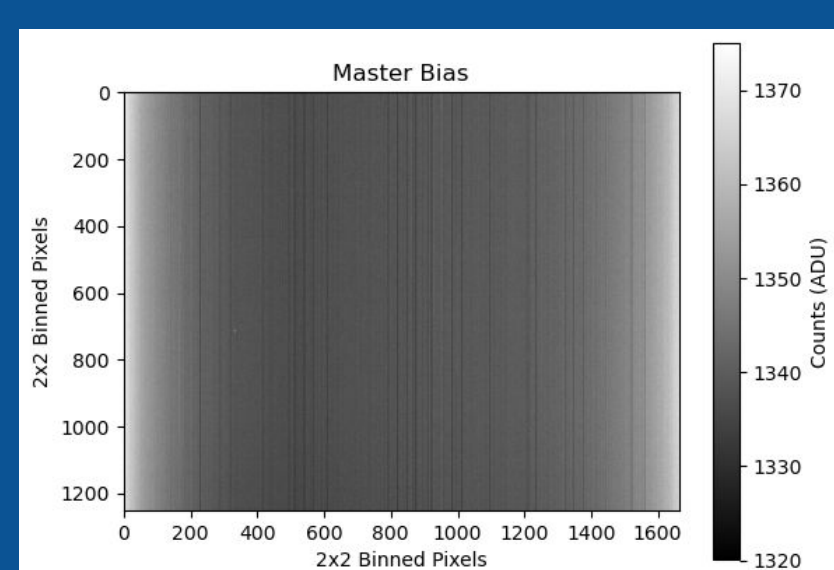
Many different factors are taken into account when selecting observation targets. Cluster type, number of stars, brightness, location, air mass, and size are all important factors that we considered. We chose to observe two globular clusters, M2 and M15, which are old and dense star clusters that contain ~100,000 or more stars.

We collected observations of M15 and M2 on 09/28/2025 and 10/05/2025, respectively, in multiple color filters (g, r, i). The telescope we used is a Cassegrain reflector, which has both a primary concave mirror and a secondary convex mirror. This design allows the telescope to be much more compact.



Data Reduction

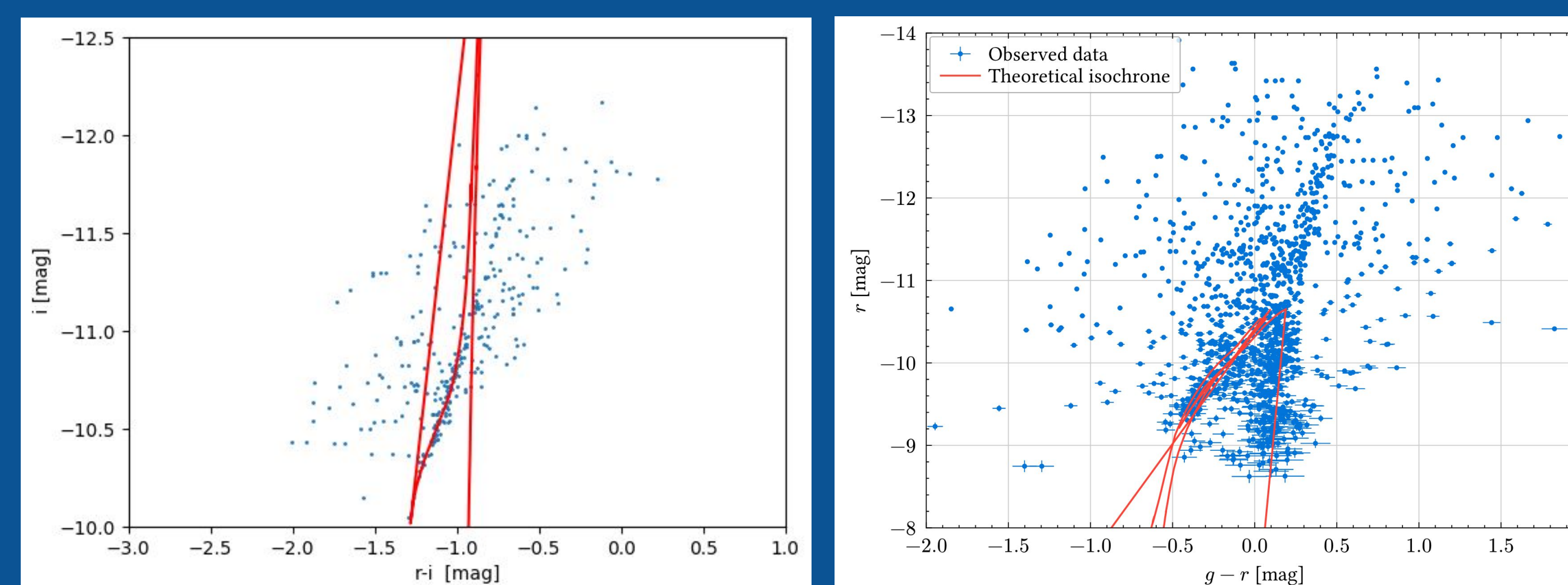
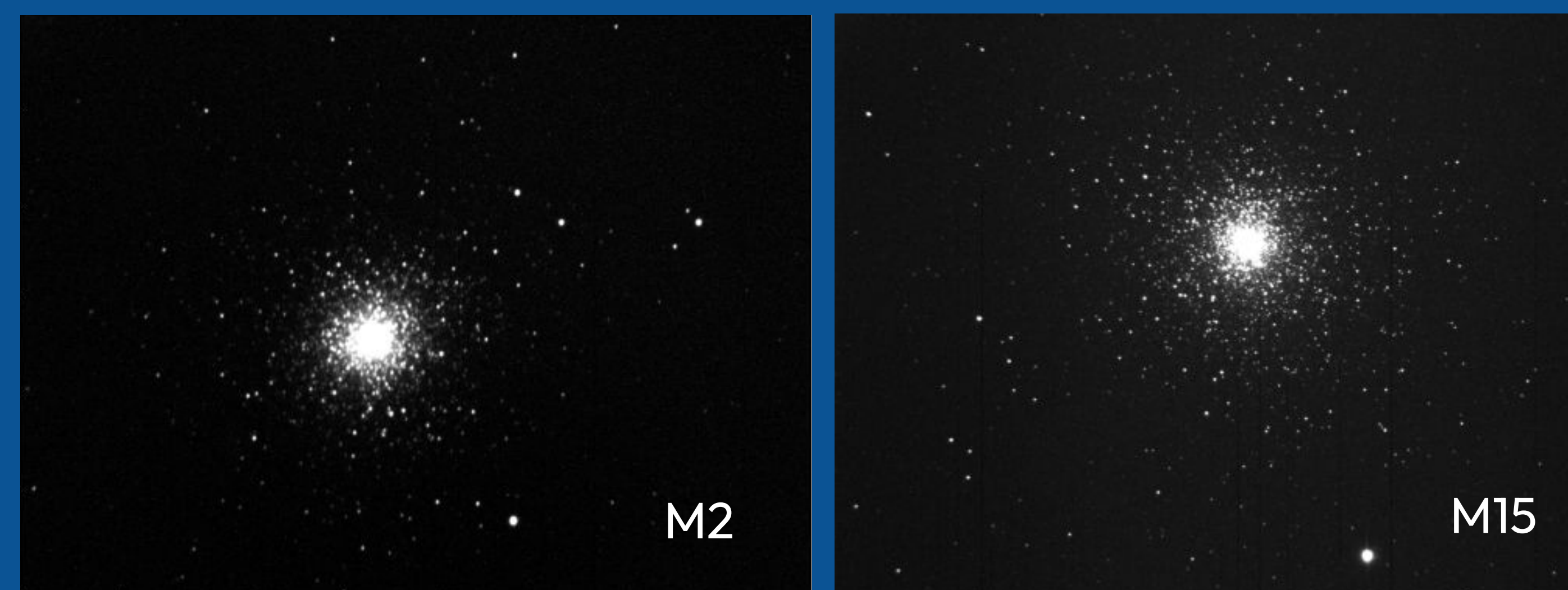
Before taking images of M2 and M15, several calibration images were taken to characterize the noise and visual interference of the telescope's CCD. Flat-field and bias images were collected and averaged into "master" bias and flat-fields, as seen below. Using the bias images, we calculated the noise emitted by the CCD. Cooling the CCD to low temperatures reduces bias noise and allows us to take clear images. Artifacts like vignetting, dust shadows, vertical bands and brightness gradients are all visible in the master calibration frames. By subtracting these factors from our raw M15 and M2 images, we ensure the cleanest possible data for further analysis.



Cluster Analysis

In both clusters, we built independent catalogs of stars and their brightnesses in the *g*, *r*, and *i* filters. We then corrected for any pointing differences between images by matching each star entry from one catalog to an entry in another. By subtracting the brightness of a matched star in one filter from its brightness in another filter, we calculate the 'color index', a vital quantity for estimating the star's age.

Plotting the color index against the brightness for each star of the cluster gives us a statistical view of the evolutionary stage of the cluster as a whole. This plot is known as a color-magnitude diagram (CMD).



Color-magnitude diagrams of M2 (left) and M15 (right). Each plot shows matched sources (blue points) and the best-fitting isochrone (red) for the given cluster.

To quantitatively estimate the properties of each cluster's stars, we matched our data to curves of constant age known as *isochrones*. These give the expected trajectory a star will evolve along from birth to death. Every star located on a given isochrone has the same age, letting us estimate the age of an entire cluster by determining the best-fitting isochrone.

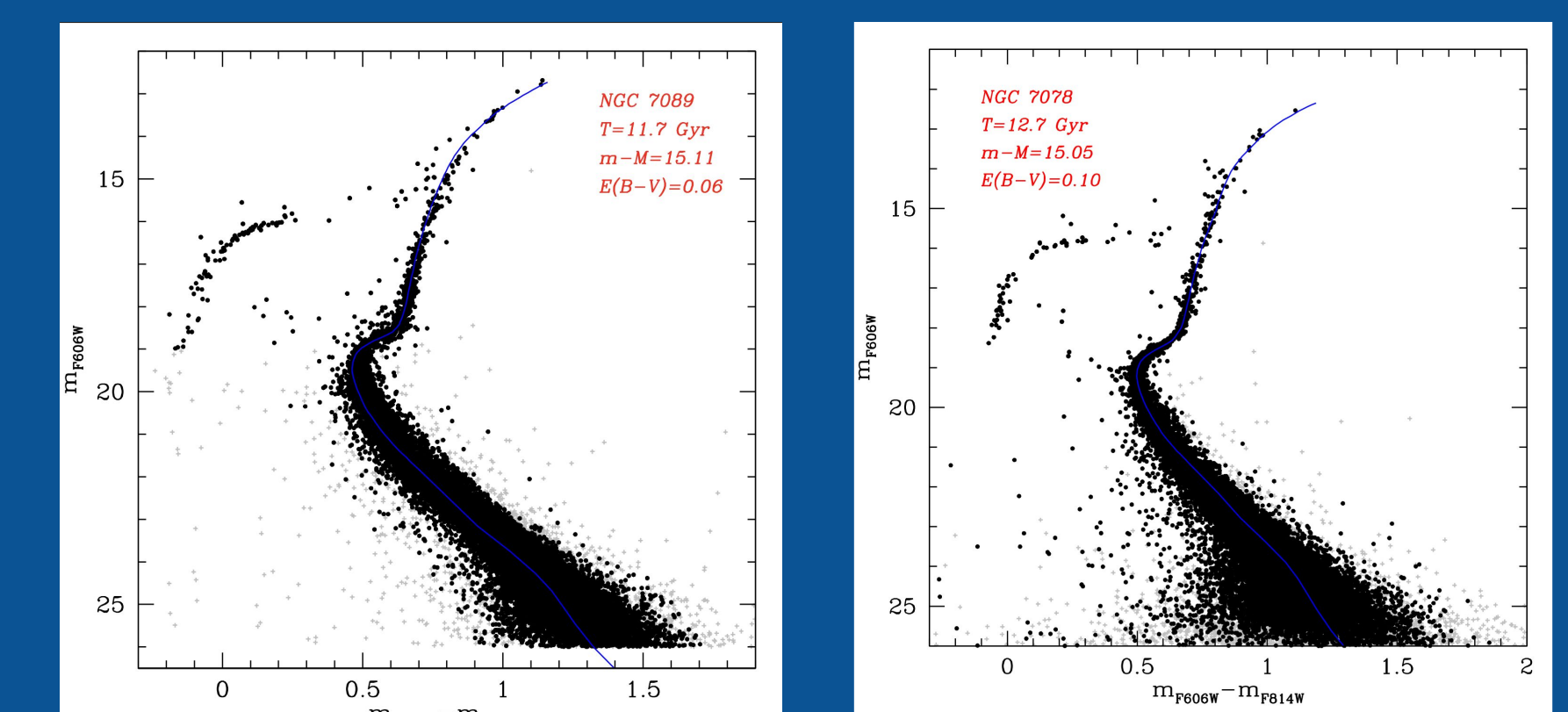
Isochrone shapes are also affected by *metallicity*, which describes the fraction of elements heavier than hydrogen or helium in a star. For M2 (left), we estimated an age of 12.589 Gyr at a metallicity of 0.03. For M15 (right), we estimated an age of 12.5 Gyr at a metallicity of 0.01.

Conclusions

The best fit isochrones to M2 and M15 indicate ages of 12.589 Gyr and 12.5 Gyr, respectively. The Holger Baumgardt cluster database reports ages of 11.5 Gyr and 12.7 Gyr for M2 and M15, respectively.

One possible explanation for this difference is that we did not differentiate between sources in the cluster and those in the foreground/background. Therefore, it is possible that stars outside the cluster biased our isochrone fitting. Another explanation is that these clusters are both relatively dense, so it is likely that sources towards the center of the cluster overlapped and counted as a single source. Finally, our detection methods were biased towards bright stars, which correspond to the upper end of the CMD. This limited the available data for isochrone fitting, which can be seen in the difference between our CMDs and the Holger Baumgardt data.

Nevertheless, these observations demonstrate the effectiveness of the Livingston-Stark Observatory in collecting accurate science data.



Color-magnitude diagrams of M2 (left) and M15 (right). Data retrieved from Baumgardt et al. "Galactic Globular Cluster Database" (2023)

References

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